

ESP Failure Analysis & Reliability for Step Change Improvement

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Jeff Dwiggins

Artificial Lift Solutions Pte Ltd



Outline

- This presentation will briefly discuss **Failure Analysis and Reliability** to implement step change improvement

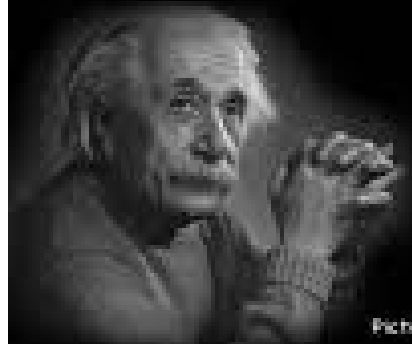
The “Genesis” of this Topic

- Petroleum Development Oman (PDO)



Words of Wisdom

**Insanity: doing the
same thing over and
over again and
expecting different
results.**



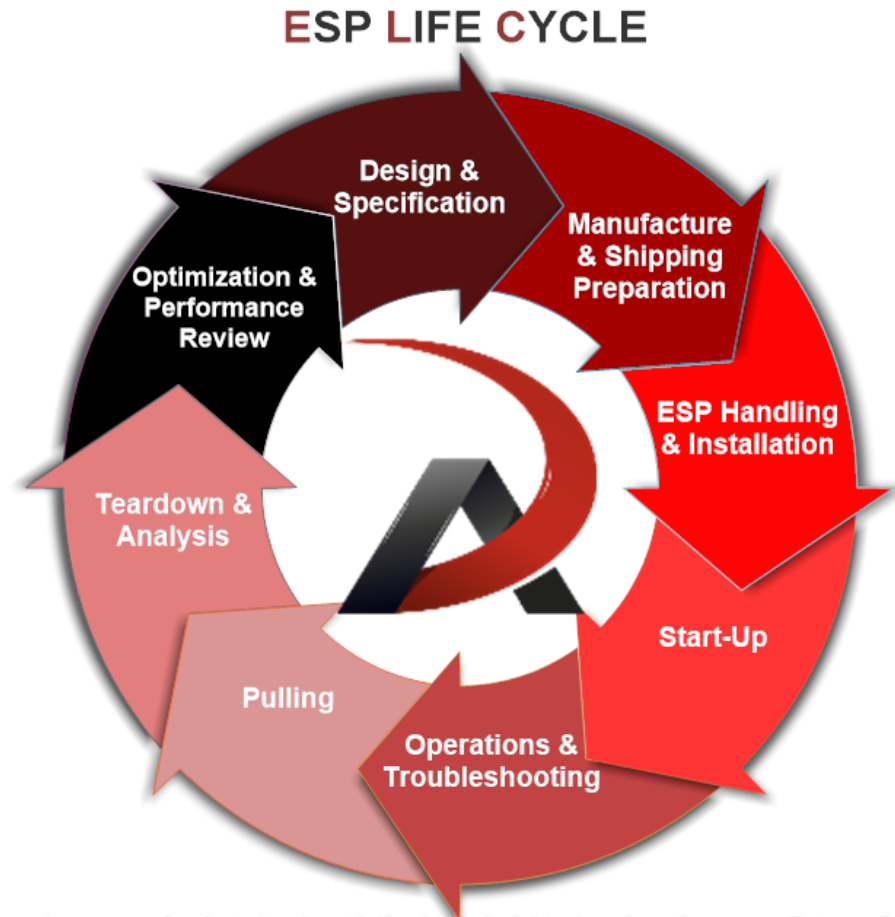
Albert Einstein

The ESP Industry

- The ESP industry is littered with failures that are repetitive
- Many (maybe even most) are avoidable
- ESP reputation is “high-volume” – but not always reliable
- Some operators will sacrifice production because of this reputation
- So where do we do things go wrong?

Life Cycle

- Most people have seen this life cycle circle before.
- You are participating willingly, or unwillingly



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A Few Common Misconceptions

1. All ESPs are the same
2. Top quality will be delivered
3. Failure analysis will be objective and without bias
4. Short cuts will not be taken

Common Deficiencies

Often, operators fail to recognize the value of focused attention on ESP quality and reliability. Often, focus only occurs after a failure(s).

1. Operators do not possess the in-house skill set
 - a) Failure to follow proper/established protocols
 - b) Failure to follow protocols leads to loss of information and therefore the ability to improve
2. When someone is the “designated” ESP person, they may not be qualified in all areas of the life cycle
 - a) E.g.; may understand ESP applications, but are not qualified as manufacturing and reliability expert
3. Assume that the goals are aligned between operator and supplier

Common Deficiencies

4. The nuances, or differences, between suppliers can be significant
5. Third party supplied items often are overlooked
6. Repeated failures (don't learn from a previous mistake)
7. Inability to have an overview of strengths and weakness between suppliers/equipment

Examples of Repeat Failure

Motor Leads Problem



Shaft Fretting



Examples of Repeat Failure

Spinning Diffusers



Scaling



Examples of Repeat Failures

Collapsed Diffuser



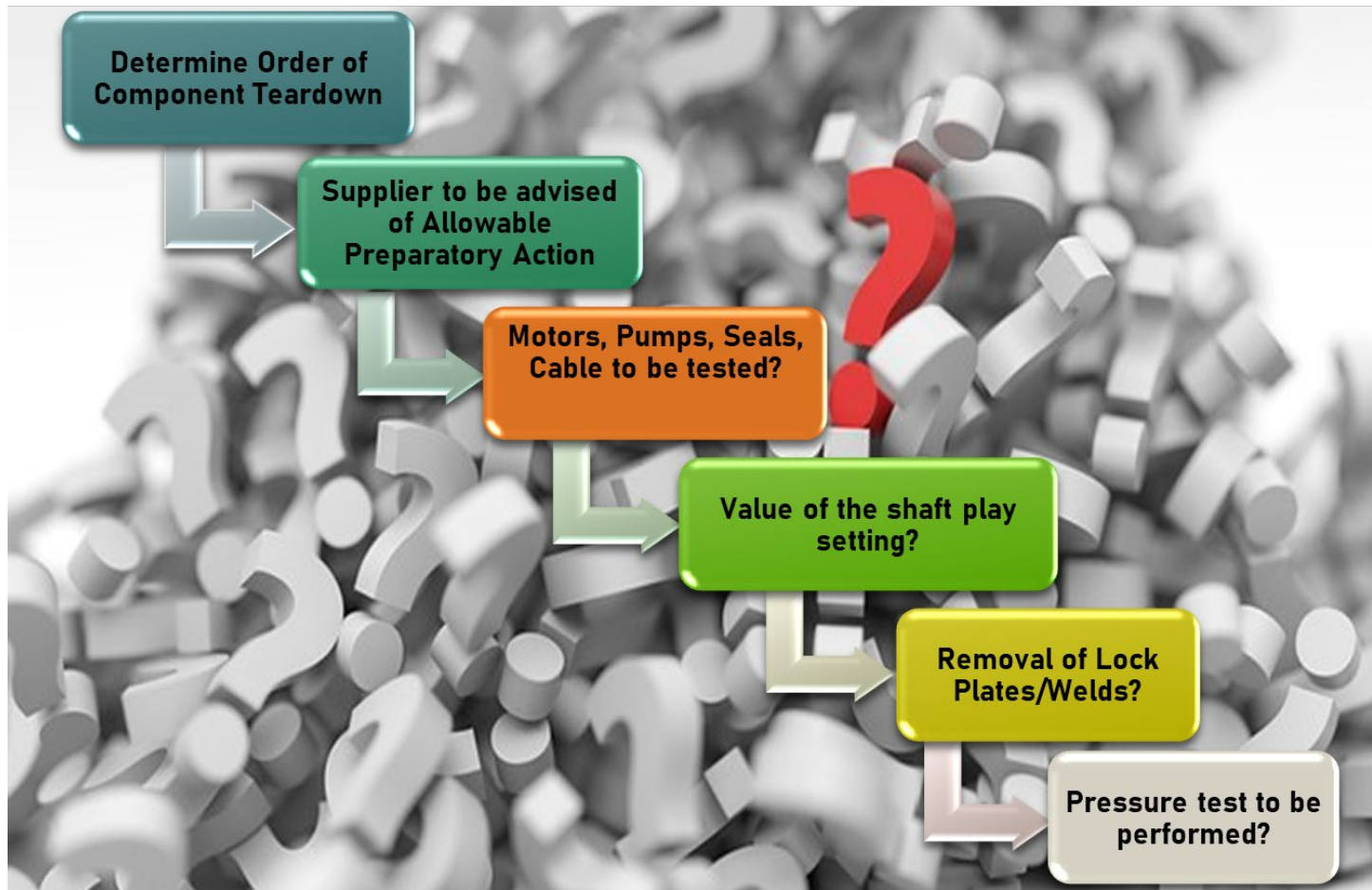
Cracked Carbide Bearings



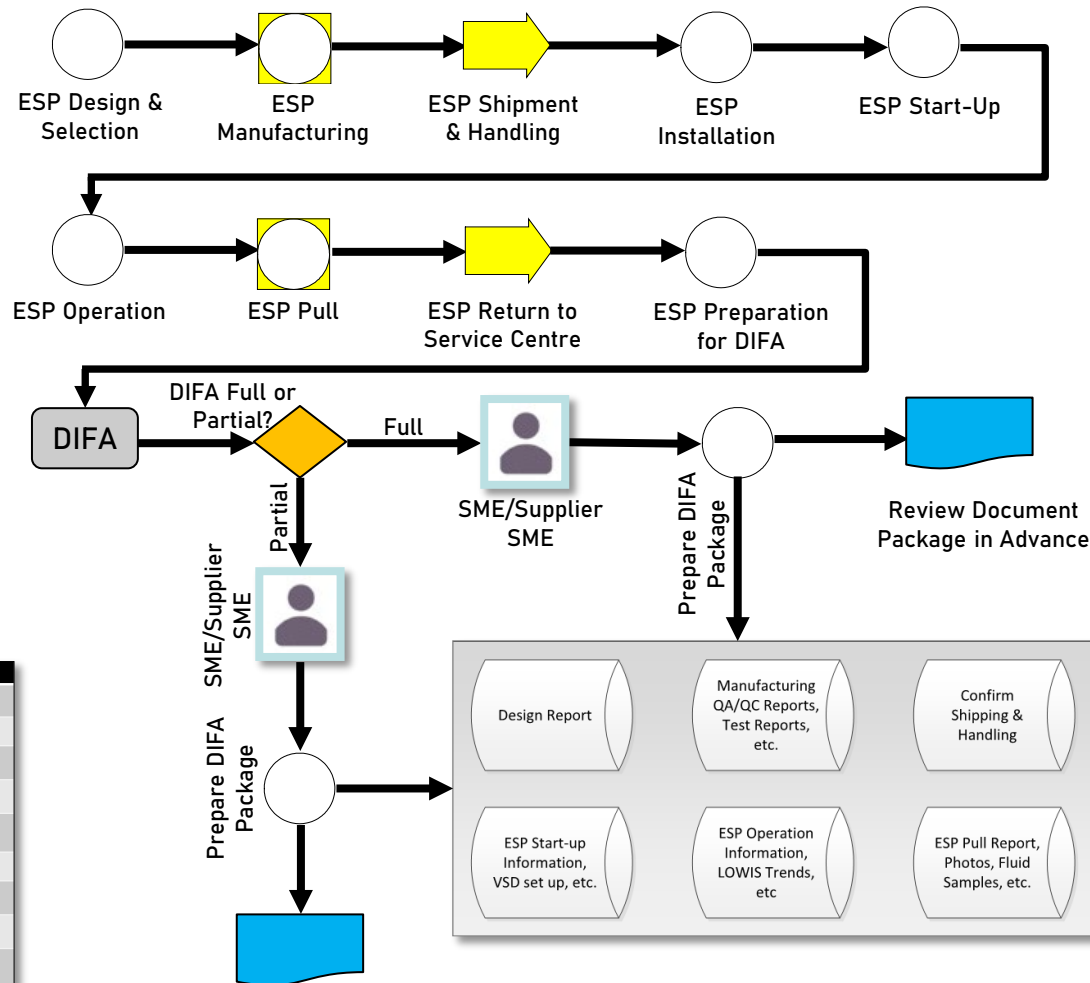
Options?

1. Get better
2. Accept the cost of failure
3. Move to another AL type and absorb the cost of that decision

Failure Analysis Methodology

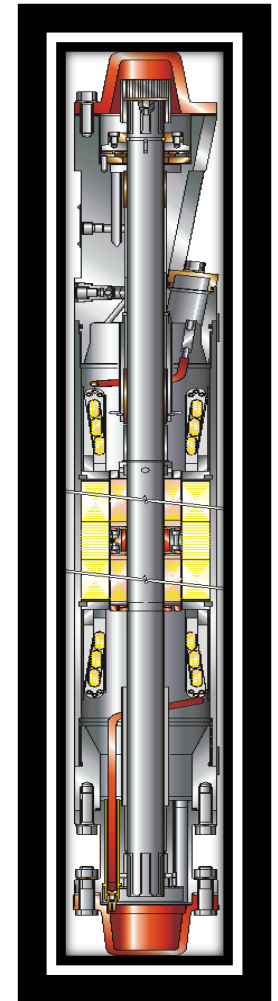
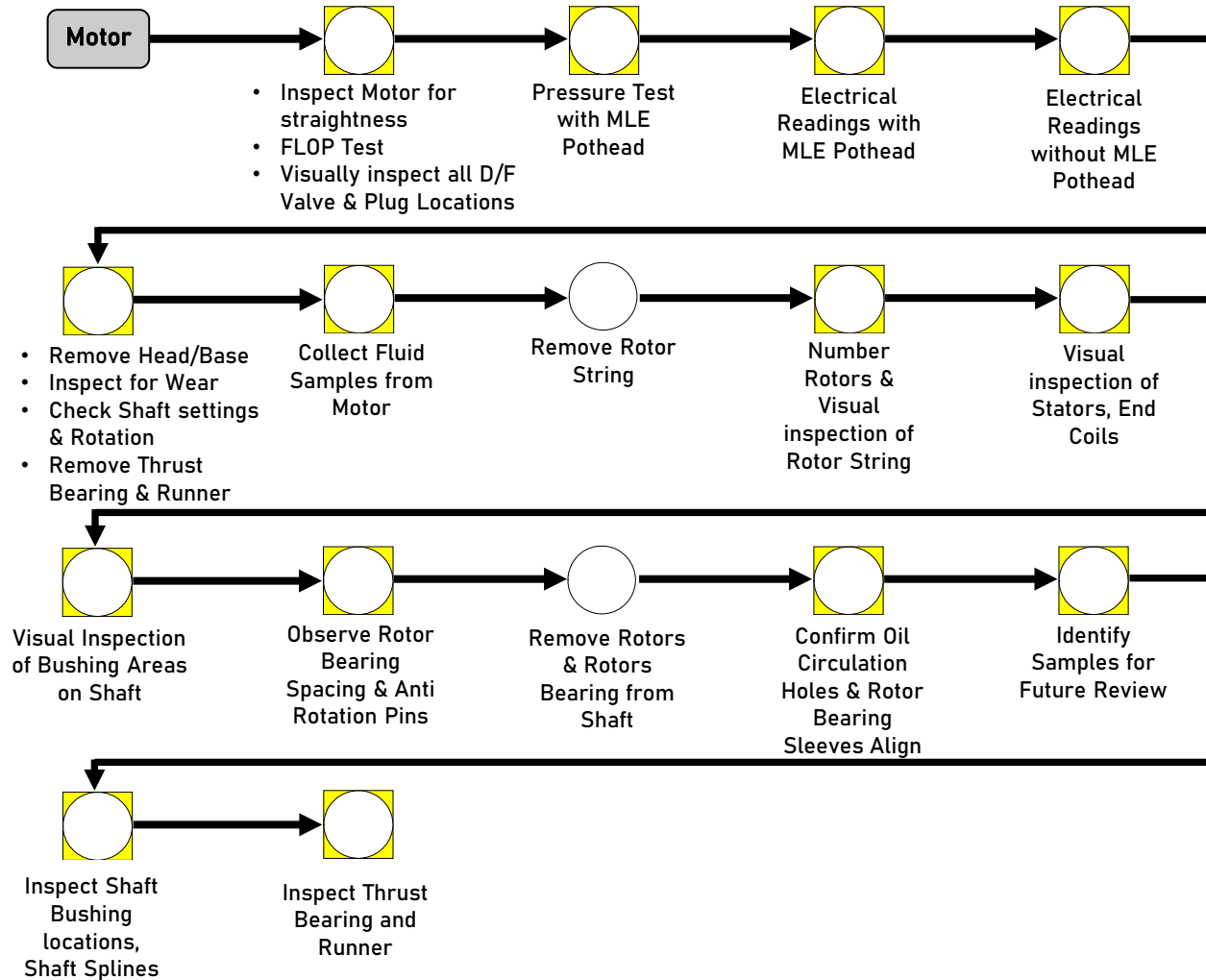


DIFA Flow Diagram - Overview

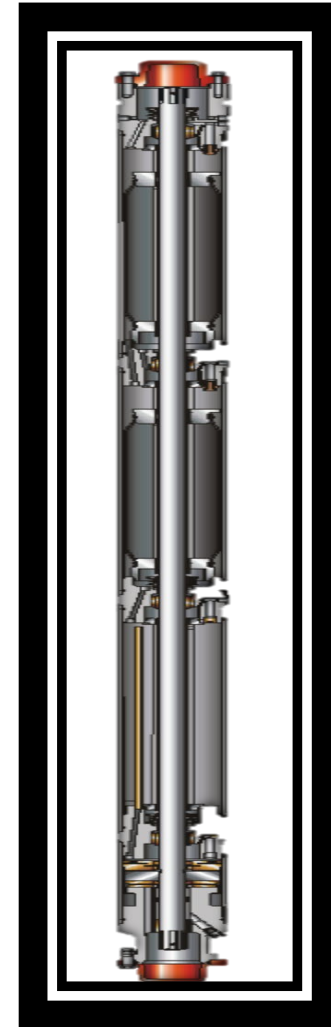
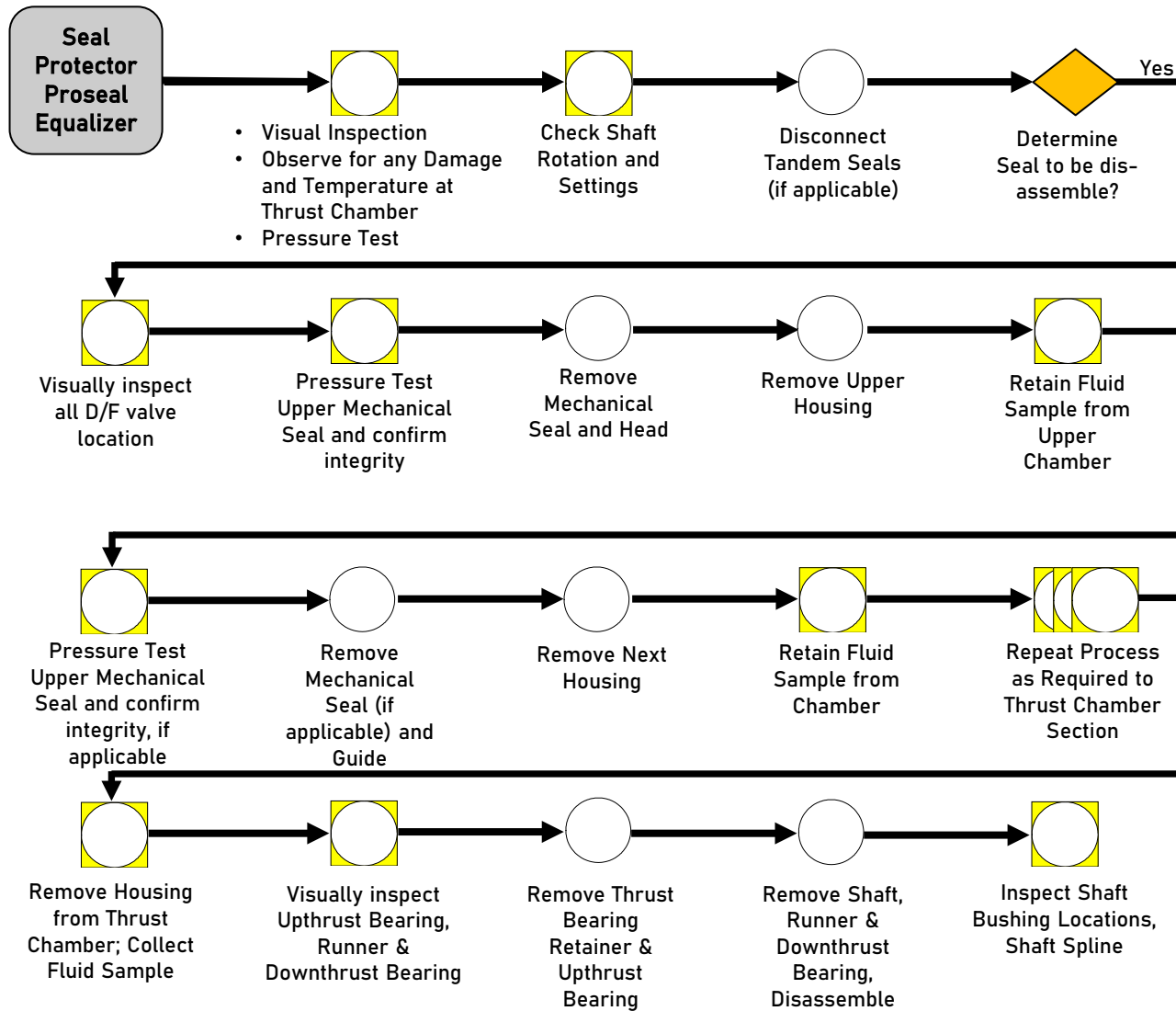


Symbols	Description
	Process Step or Operation (White)
	Inspection (Yellow)
	Handling, Transport or Movement of Material (Yellow)
	Process and Inspection with Recording (White and Yellow)
	Input, Output and Process Name
	Start or Ending of a Process (Turquoise)
	Decision between two or more points (Orange)
	Subject Matter Expert
	Document (Blue)

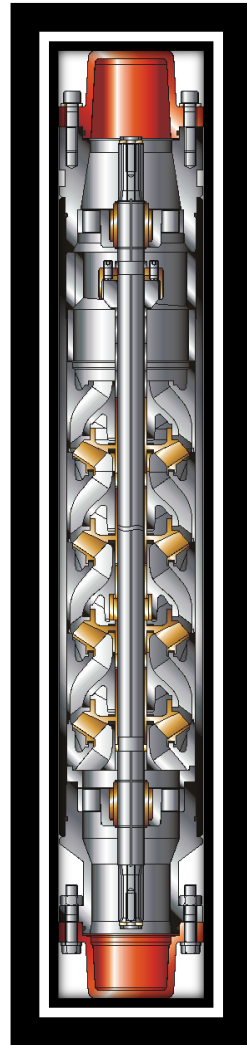
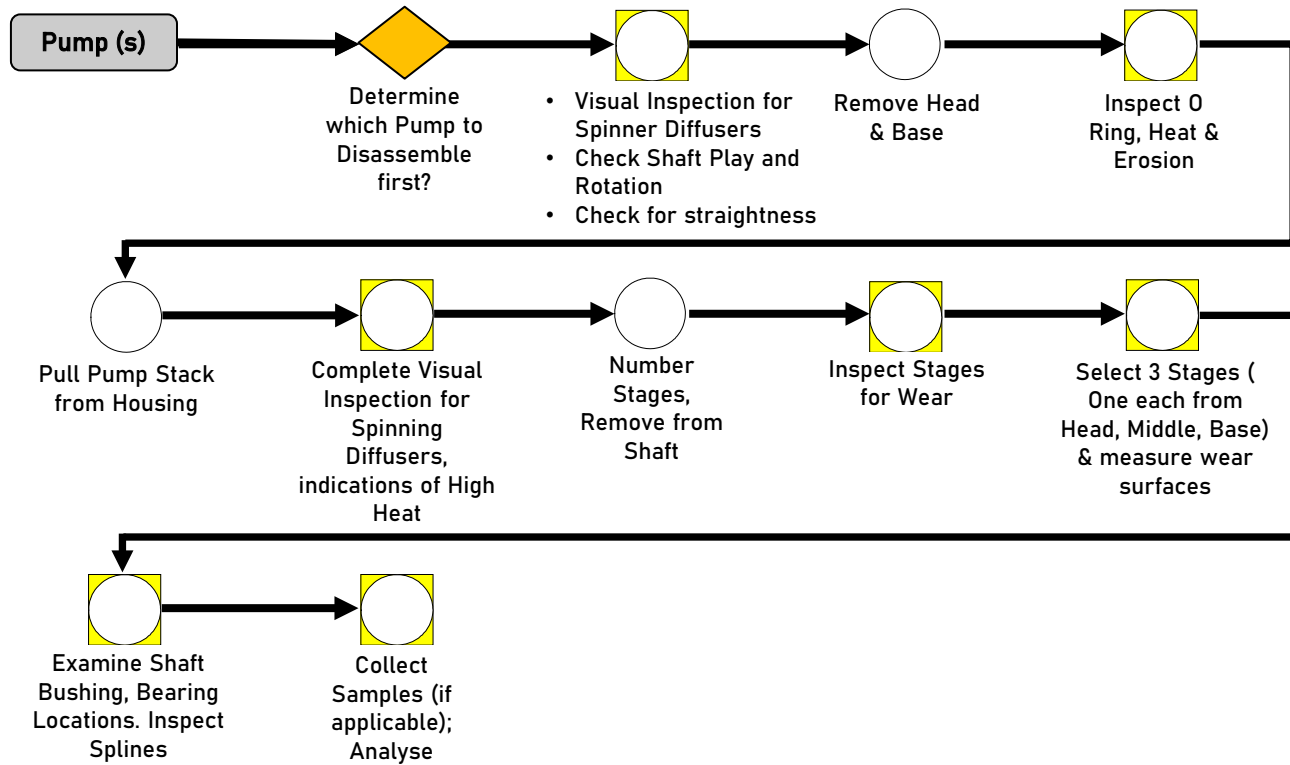
MOTOR DIFA FLOW CHART



SEAL DIFA FLOW CHART



PUMP DIFA FLOW CHART



Technology

- The DIFA process remains manpower intensive in the majority of cases
- An effort is underway to digitize this data and utilize technology
- A new “DIFA” app is being utilized

Nomenclature

- There is a tendency to confuse, or complicate, our efforts based upon nomenclature
- Our descriptions can be subjective (i.e., moderate versus heavy, etc.)
- A new API RP 11S1 (currently under development) will give guidance on nomenclature, etc.

For Example.....

Nomenclature Challenge

Head	Pull & Run Report	Installation Report	POOH Report	Pump Efficiency	Compression
Modified Compression	Floater	Modular	Center Tandem	Upper Tandem	Lower Tandem
Middle Tandem	Axial Flow	Pancake	Mixed Flow	Radial Flow	Balance Hole
Balance Ring	Down-thrust washer	Up-thrust washer	Stage balancing	Bearing Housing	Bearing Support
Wagon Wheel	Break Water	Anti-Swirl Ribs	Sand Dams	Compression Tube	Compression Nut
Lower Shroud	Upper Shroud	Eye Opening	Impeller	Diffuser	Pump Stage
Pump Series Descript	Bag Chamber	Labyrinth Chamber	Babbit Bearing	High Load Bearing	Tilt Pad Bearing
Coil Forming Block	Kapton	Polyimide	Phase Barrier	Tape and Braid	Extruded Barrier
Encapsulated	Varnished	Coil Supports	Cookie Cutter	Motor Lead Adapter	Pothole
Pothole	Plug In	Tape In	Manufacturing Quality	Wellbore Conditions	Operating Error
Handling Error	Installation Error	Infant Failure	Rotor Bearing	Rotor	Anti-Rotation Clip
T-Rings	BOM	BHT	BHP	Vortex Gas Separator	Rotary Gas Separator
AGH	GINPSHL	Vapro	Integral Intake	Brush Cable	Motor Leads
D&F	Coupling	Two-Piece Coupling	Bolt on Intake	Adaptors	Stator Laminations
Lams	Up Tube	Down Tube	Shaft Tube	UMB	Gauge Ready Base
FCE	MLE	Flop	GOR	G0-No-Go	SSHB
HSS	Inconel	Lock Plates	Lock Welds	Mag Wire	Extruded Barriers
O-ring Boss	O-ring Groove	Micro finish	Tool Mark Chatter	Lapping	Smearing
Loss of Lubricity	TDR	TDH	TIR	THD	Spline
Slinger	Pitting	Pigtail	EFT	Feed-through	RIH
VSD	VFD	SWBD	MLE Splice	Lower Pigtail Splice	Field Attachable
PODs	Y-Tool	By-Pass	Sand Guard	Discharge Head	Discharge Pressure Sub
Mushroom Sleeve	T-Sleeve	Spacer	Key Stock	Tungsten Carbide	Silicon Carbide
SC	WC	TC	Shedder	Ni-resist	Type 4
Surface Switchgear	Transformer	XFRMR	Flex	Seal	Protector
Pro-Seal	Equalizer	Continuum	MGH	Maximus	Gauge
Sensor	Radial Play	Axial Play	Shaft Setting	Shaft Play	Resistance to Ground
Phase to Phase	EPDM	CELF	J-Box	Protectorliser	CCP
Lasalle Clamp	Cast Clamp	Stamped Clamp	MLE Clamp	Bands	Optimisation
Voltage Optimisation	Trend Analysis	Slot Liner	Thrust Chamber	#4 Flat Lead	#2 Flat Lead

Closing

1. Conducting failure analysis requires focused attention from “someone” – PDO’s effort is the best example of focused improvement
2. Detailed and repeatable methods should be applied
3. Suppliers and Operators do not always share the same goals
4. Proper training and utilization of technology is imperative.
5. Proper focus on all aspects of the life cycle should produce improvement for all parties